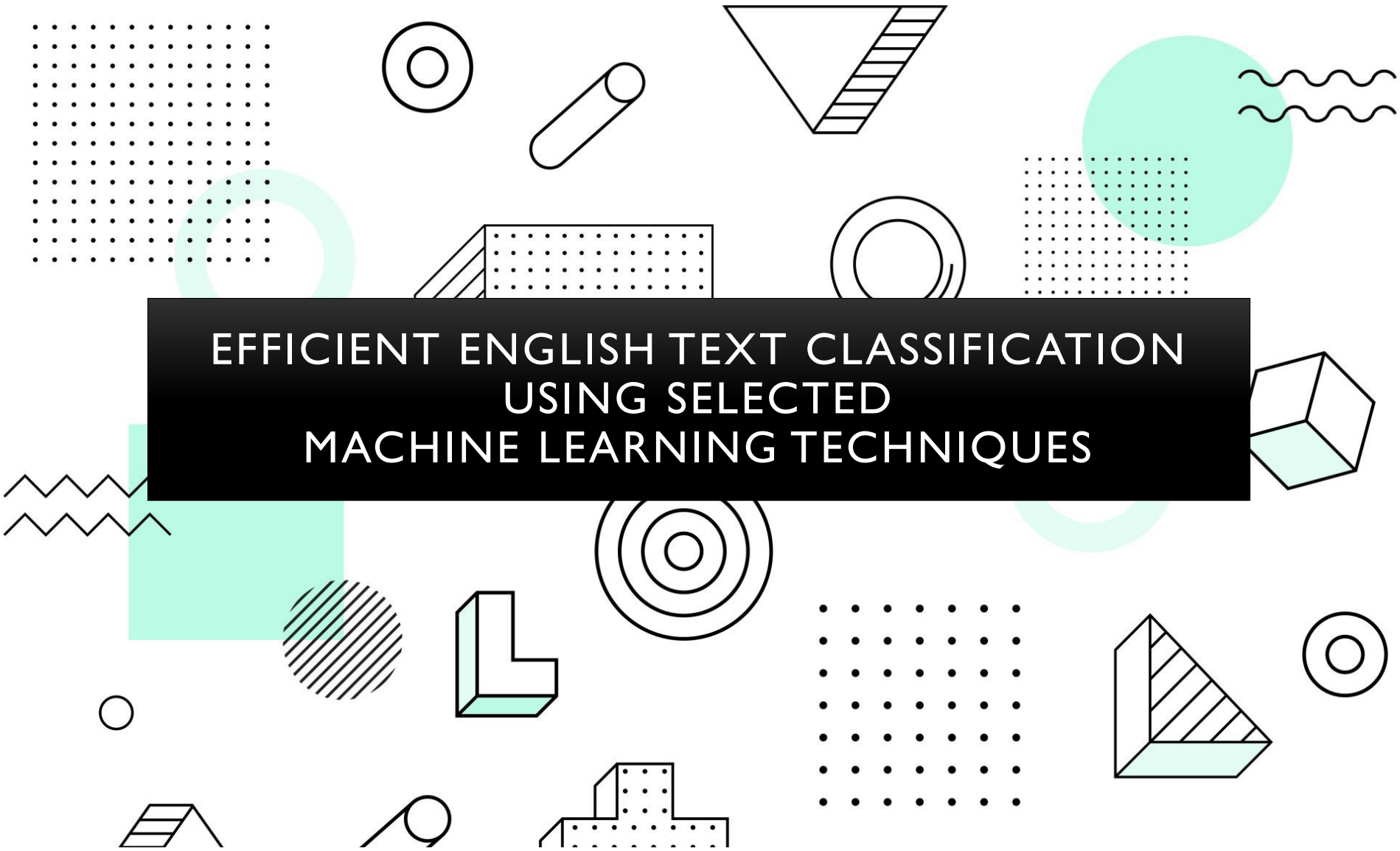
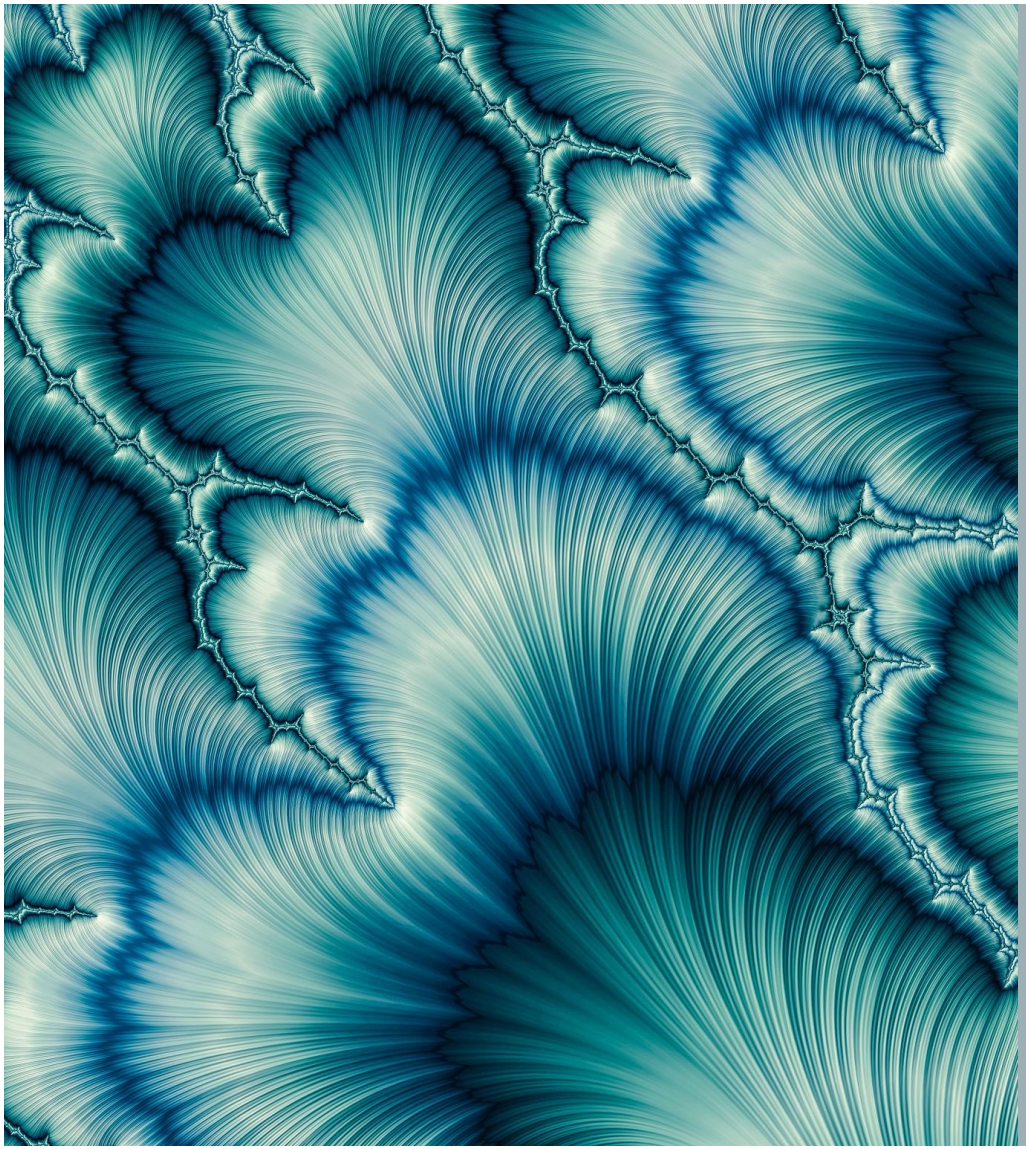

EFFICIENT ENGLISH TEXT CLASSIFICATION
USING SELECTED
MACHINE LEARNING TECHNIQUES



INTRODUCTION

TITLE AND OBJECTIVE

- **Title**: Efficient English Text Classification using Selected Machine Learning Techniques
- **Authors**: Xiaoyu Luo (Hunan University of Technology and Business, China)
- **Journal & Publication Date**: *Alexandria Engineering Journal*, February 2021
- **Objective of the Study**:
  - To identify efficient machine learning techniques for English text classification by comparing the performance of various classifiers.
  - A specific focus on Support Vector Machines (SVM) to evaluate its effectiveness relative to other models like Logistic Regression and Naïve Bayes.

INTRODUCTION - SIGNIFICANCE OF TEXT CLASSIFICATION

- **Text Classification (TC)**:
  - Technique for categorizing documents into predefined classes using machine learning.
  - Useful for organizing vast amounts of unstructured data for better retrieval and management.
- **Growing Demand**:
  - Significant growth in unstructured text data from corporate records, government, and personal communications.
  - Increase in blogging, social media, and online interactions has amplified the need for efficient classification methods.

MACHINE LEARNING FOR TEXT CLASSIFICATION

- **Role of ML in Text Classification**:
  - **Supervised Learning**: Uses labeled data for training, ensuring accurate categorization.
  - **Preprocessing**: Involves stop-word removal, stemming, and feature selection for optimized performance.
- **Popular Applications**:
  - **Spam Filtering**: Recognizes and filters unwanted emails.
  - **Sentiment Analysis**: Detects opinions on products and services.
  - **Fake News Detection**: Identifies fraudulent or counterfeit content online.

KEY CHALLENGES

Challenges in Efficient Text Classification:

- **Misclassification**:
  - Handling incorrectly labeled data remains challenging, especially in domains where language or context may lead to ambiguities.
- **Feature Selection and Dimensionality Reduction**:
  - With thousands of potential features (words or terms), feature selection is critical to manage computational load and enhance classifier performance.
  - Selecting the right features without compromising classification accuracy is essential.

MACHINE LEARNING ALGORITHMS

SVM (SUPPORT VECTOR MACHINE)

Overview ☞ SVM is a classification algorithm that finds the optimal hyperplane to separate data into classes.

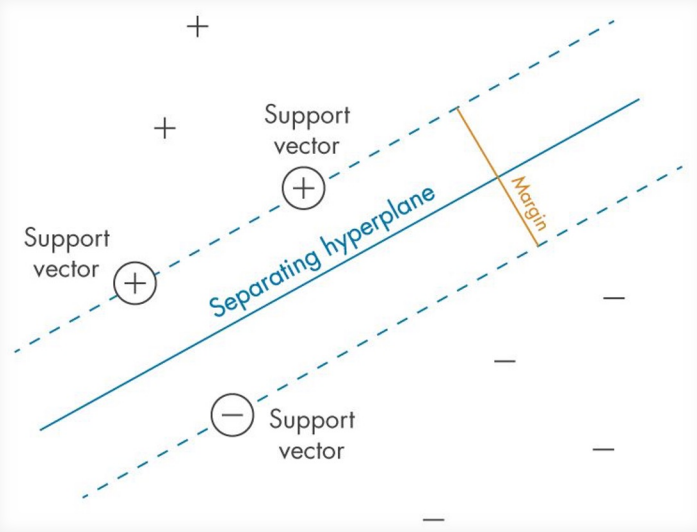
Key Concepts:

- **Hyperplane**: Divides data into different classes.
- **Support Vectors**: Key data points that influence the hyperplane's position.
- **Maximal Margin**: SVM maximizes the margin between classes to improve classification accuracy.

Advantages:

- Effective in high-dimensional data, ideal for text classification.
- Reduces overfitting with a clear margin of separation.

Applications: Used in spam detection, sentiment analysis, and document categorization.



NAÏVE BAYES (NB)

Overview: NB is a probabilistic classifier that applies Bayes' theorem with the assumption of independence between features.

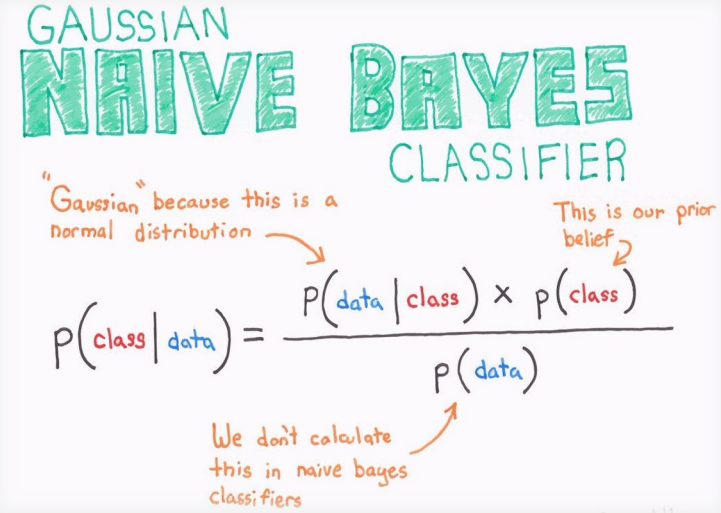
Key Concepts:

- **Conditional Probability**: Calculates the probability of a class given the input features.
- **Feature Independence**: Assumes each feature contributes independently to the probability.

Advantages:

- Simple, efficient, and works well with large datasets.
- Ideal for high-speed text classification tasks.

Applications: Used in spam filtering, document categorization, and sentiment analysis.

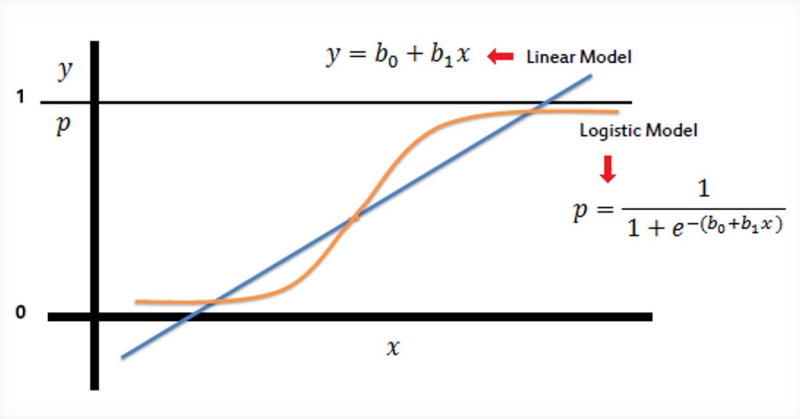


LOGISTIC REGRESSION (LR)

Overview: LR is a linear model for binary classification, predicting probabilities and classifying based on a threshold.

Key Concepts:

- **Sigmoid Function**: Maps inputs to a probability between 0 and 1.
- **Decision Boundary**: Separates data into two classes based on a threshold.



Advantages:

- Interpretable model with efficient performance in binary tasks.
- Suitable for text data with linear separability.

Applications: Used in sentiment analysis, binary text classification, and medical diagnosis.

METHODOLOGY AND IMPLEMENTATION SETUP

IMPORTING NECESSARY LIBRARIES

- Basic: Numpy, Pandas
- Dataset: fetch_20newsgropus from sklearn
- Model: SVC, MultinomialNB, LogisticRegression from sklearn
- Metrics: sklearn libraries
- Plot: matplotlib

```
# Import necessary libraries
import pandas as pd
import numpy as np
from sklearn.datasets import fetch_20newsgroups
from sklearn.model_selection import train_test_split, GridSearchCV, cross_val_score
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.svm import SVC
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
import matplotlib.pyplot as plt
import nltk
import re
```

PREPROCESSING ON TEXT

Data Preprocessing

- Steps taken: Tokenization, stop-word removal, and stemming (PorterStemmer)
- Benefits of preprocessing: Improves feature relevance, reduces dimensionality, and optimizes performance.
- Library: nltk

```
# Download NLTK stopwords
nltk.download('stopwords')
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

# Initialize stop words and stemmer
stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()

# Preprocessing function
def preprocess_text(text):
    text = re.sub(r'\W', ' ', text)  # Remove non-alphanumeric characters
    text = text.lower()  # Convert to lowercase
    words = [stemmer.stem(word) for word in text.split() if word not in stop_words]
    return ' '.join(words)
```

PREPARING DATA

Dataset Description

•Data sources: UCI library and other English news websites.

•Overview of dataset categories: Topics like sports, literature, campus news, etc.

Prepared three categories of data as described in paper.

```
# Fetch 20 Newsgroups data and divide into 3 datasets to simulate different topic groups
categories1 = ['alt.atheism','comp.graphics','comp.os.ms-windows.misc','comp.sys.ibm.pc.hardware','comp.sys.mac.hardware','comp.windows.x','misc.forsale']
categories2 = ['rec.autos','rec.motorcycles','rec.sport.baseball','rec.sport.hockey','sci.crypt','sci.electronics','sci.med']
categories3 = ['sci.space','soc.religion.christian', 'talk.politics.guns', 'talk.politics.mideast', 'talk.politics.misc', 'talk.religion.misc']

# Load different subsets
data1 = fetch_20newsgroups(subset='all', categories=categories1, shuffle=True, random_state=42)
data2 = fetch_20newsgroups(subset='all', categories=categories2, shuffle=True, random_state=42)
data3 = fetch_20newsgroups(subset='all', categories=categories3, shuffle=True, random_state=42)

# Create DataFrames for each simulated dataset and preprocess text
df1 = pd.DataFrame({'text': [preprocess_text(text) for text in data1.data], 'category': data1.target})
df2 = pd.DataFrame({'text': [preprocess_text(text) for text in data2.data], 'category': data2.target})
df3 = pd.DataFrame({'text': [preprocess_text(text) for text in data3.data], 'category': data3.target})
```

MODEL TRAINING

Steps:

- Vectorize the text data
- Split the train and the test data
- Define the parameters for hyperparameter tuning

```python
# Define function to train and evaluate models with hyperparameter tuning
def train_and_evaluate_models(df, dataset_name):
    # Vectorize the text data
    vectorizer = CountVectorizer(max_features=4000, stop_words='english')
    X = vectorizer.fit_transform(df['text']).toarray()
    y = df['category']

    # Split data into training and testing sets
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

    # Define parameter grids for hyperparameter tuning
    param_grid_svm = {
        'C': [0.1, 1, 10],
        'kernel': ['linear', 'rbf'],
        'gamma': ['scale', 'auto']
    }
    param_grid_nb = {'alpha': [0.1, 0.5, 1.0, 2.0]}
    param_grid_lr = {'C': [0.1, 1, 10], 'solver': ['liblinear', 'lbfgs']}
```

MODEL TRAINING

Steps:

- Model initialization using GridSearchCV
- Fit the models
- Get the best estimators

```
# Initialize models with GridSearchCV
grid_search_svm = GridSearchCV(SVC(), param_grid_svm, cv=5, scoring='accuracy', n_jobs=-1)
grid_search_nb = GridSearchCV(MultinomialNB(), param_grid_nb, cv=5, scoring='accuracy', n_jobs=-1)
grid_search_lr = GridSearchCV(LogisticRegression(max_iter=200), param_grid_lr, cv=5, scoring='accuracy', n_jobs=-1)

# Fit models
grid_search_svm.fit(X_train, y_train)
grid_search_nb.fit(X_train, y_train)
grid_search_lr.fit(X_train, y_train)

# Get best estimators
best_svm = grid_search_svm.best_estimator_
best_nb = grid_search_nb.best_estimator_
best_lr = grid_search_lr.best_estimator_
```

MODEL TRAINING

Steps:

- Each model evaluation on test data
- Store the result metrics for further comparison
- Return the result

```python
# Evaluate each model on the test data
y_pred_svm = best_svm.predict(X_test)
y_pred_nb = best_nb.predict(X_test)
y_pred_lr = best_lr.predict(X_test)

# Store metrics
results = {
    f'{dataset_name} - SVM': {
        'Accuracy': accuracy_score(y_test, y_pred_svm),
        'Precision': precision_score(y_test, y_pred_svm, average='macro'),
        'Recall': recall_score(y_test, y_pred_svm, average='macro'),
        'F1 Score': f1_score(y_test, y_pred_svm, average='macro')
    },
    f'{dataset_name} - Naive Bayes': {
        'Accuracy': accuracy_score(y_test, y_pred_nb),
        'Precision': precision_score(y_test, y_pred_nb, average='macro'),
        'Recall': recall_score(y_test, y_pred_nb, average='macro'),
        'F1 Score': f1_score(y_test, y_pred_nb, average='macro')
    },
    f'{dataset_name} - Logistic Regression': {
        'Accuracy': accuracy_score(y_test, y_pred_lr),
        'Precision': precision_score(y_test, y_pred_lr, average='macro'),
        'Recall': recall_score(y_test, y_pred_lr, average='macro'),
        'F1 Score': f1_score(y_test, y_pred_lr, average='macro')
    }
}

return results
```

GETTING THE RESULT

Steps:

- Find the result for each model for each data
- Combine and plot the graph of the result.

```python
# Train and evaluate models for each dataset
results1 = train_and_evaluate_models(df1, "Data1")
results2 = train_and_evaluate_models(df2, "Data2")
results3 = train_and_evaluate_models(df3, "Data3")

# Combine all results
all_results = {**results1, **results2, **results3}
results_df = pd.DataFrame(all_results).T
print("Model Comparison Results:\n", results_df)

# Plotting the results for comparison
metrics = ['Accuracy', 'Precision', 'Recall', 'F1 Score']
fig, axs = plt.subplots(2, 2, figsize=(16, 10))
fig.suptitle("Performance Comparison of Models Across Datasets", fontsize=16)

# Plot each metric
for i, metric in enumerate(metrics):
    ax = axs[i // 2, i % 2]
    svm_scores = results_df[results_df.index.str.contains("SVM")][metric]
    nb_scores = results_df[results_df.index.str.contains("Naive Bayes")][metric]
    lr_scores = results_df[results_df.index.str.contains("Logistic Regression")][metric]

    x = np.arange(3)  # Number of datasets (Data1, Data2, Data3)
    width = 0.25  # Width of the bars

    # Plot bars for each model type
    ax.bar(x - width, svm_scores, width, label='SVM')
    ax.bar(x, nb_scores, width, label='Naive Bayes')
    ax.bar(x + width, lr_scores, width, label='Logistic Regression')

    # Labeling details
    ax.set_xlabel('Datasets')
    ax.set_ylabel(metric)
    ax.set_title(f'{metric} Comparison')
    ax.set_xticks(x)
    ax.set_xticklabels(['Data1', 'Data2', 'Data3'])
    ax.legend()

    # Adding text labels on top of bars
    for j in range(len(x)):
        ax.text(x[j] - width, svm_scores.iloc[j] + 0.01, f'{svm_scores.iloc[j]:.2f}', ha='center')
        ax.text(x[j], nb_scores.iloc[j] + 0.01, f'{nb_scores.iloc[j]:.2f}', ha='center')
        ax.text(x[j] + width, lr_scores.iloc[j] + 0.01, f'{lr_scores.iloc[j]:.2f}', ha='center')

plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()
```

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$F_\beta = \frac{1 + \beta^2}{\frac{\beta^2}{Recall} + \frac{1}{Precision}}$$

EVALUATION METRICS

- **Evaluation Metrics**
- Metrics used: Precision, recall, F1-score, accuracy.
- Importance: These metrics provide a well-rounded view of model performance, particularly in multi-class settings.

RESULTS AND COMPARATIVE ANALYSIS

- SVM has more precision as compared to NB and LR for the datasets that we have selected.

Table 1 Comparative analysis of the machine learning algorithm.

	Data1			Data 2			Data 3		
	Precision	**Recall**	**F1 Value**	**Precision**	**Recall**	**F1 Value**	**Precision**	**Recall**	**F1 Value**
SVM	0.88	0.87	0.86	0.76	0.71	0.71	0.63	0.54	0.51
NB	0.40	0.28	0.24	0.27	0.29	0.18	0.12	0.33	0.18
LR	0.83	0.85	0.81	0.67	0.57	0.49	0.64	0.63	0.63

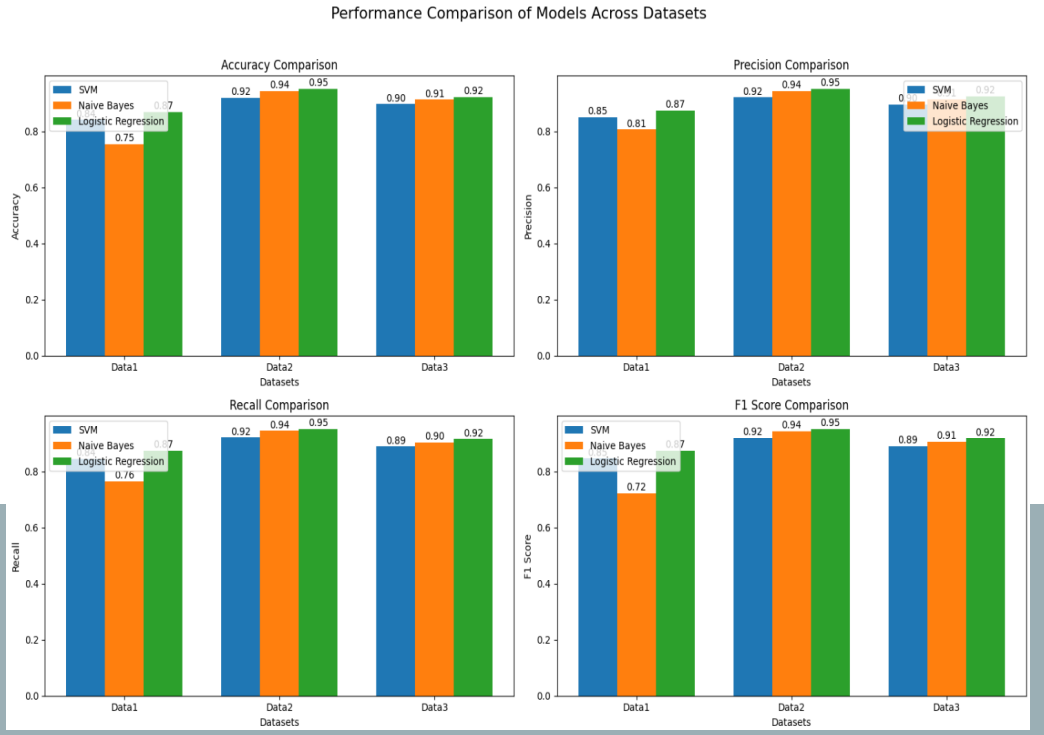
RESULTS

Insights from Comparative Analysis

- SVM shows superior performance in precision and recall like data1.
- NB effective on smaller data like data3.

Model Comparison Results:

	Accuracy	Precision	Recall	F1 Score
Data1 - SVM	0.841000	0.850862	0.844965	0.845896
Data1 - Naive Bayes	0.754000	0.805844	0.763698	0.720904
Data1 - Logistic Regression	0.870000	0.874269	0.873248	0.873435
Data2 - SVM	0.920345	0.920910	0.920301	0.919683
Data2 - Naive Bayes	0.944818	0.944046	0.944345	0.943993
Data2 - Logistic Regression	0.951056	0.950678	0.950388	0.950293
Data3 - SVM	0.896947	0.896177	0.888511	0.890665
Data3 - Naive Bayes	0.912850	0.913918	0.902964	0.905375
Data3 - Logistic Regression	0.922392	0.924298	0.915385	0.917593



DISCUSSION

- **Challenges in Text Classification**:
  - **Data Collection Impact**: Quality of data collection affects preprocessing and text mining accuracy.
  - **Preprocessing Essentials**:
  - Tokenization, stop word removal, stemming, and vector space model required for organizing data.
  - **Literature Insights**:
  - Increasing dataset size during training enhances evaluation accuracy.

CONCLUSION & FUTURE WORK

Conclusion:

- Rapid growth of text classification applications in IT.
- **SVM** outperforms NB and LR on selected datasets with higher precision, recall, and F1-score.
- Each algorithm has strengths and weaknesses depending on dataset size and characteristics.

Future Work:

- Expand classification to larger datasets, like the BBC dataset.
- Explore implementations in advanced tools: TensorFlow, Python, R, or Matlab.

BIBLIOGRAPHY

Code Link

https://colab.research.google.com/drive/1op8272F2ciF2NE7BZkHaKgTEcOzfvIeq?usp=sharing

Resources

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THANK YOU